Emergent categories in usage-based language learning

Anna Jon-And[*1,2] and Jérôme Michaud[1,3]

[*]Corresponding Author:anna.jon-and@su.se
[1]Centre for cultural evolution, Department of Psychology, Stockholm University, Stockholm, Sweden
[2]Department of Romance Studies and Classics, Stockholm University, Stockholm, Sweden
[3]Department of Business and Mathematics, Mälardalen University, Västerås, Sweden

Categories are central to linguistic productivity, yet their emergence remains poorly understood. Building on previous pilot work, we present a usage-based model in which grammatical categories arise spontaneously from minimal cognitive principles of sequence representation and chunking. The model is tested on artificial languages containing intransitive, monotransitive, ditransitive, and subordinate-clause constructions. Without any predefined categories grammatical information, the system develops emergent combinatorial formulas—*types*—that encode how linguistic elements can be combined and generalized. Learning with such emergent types is faster and more robust than learning without them, particularly as structural complexity and vocabulary size increase. The results suggest that grammatical categories can evolve as an adaptive response to the combinatorial challenge of language learning, providing a formal account of how construction-like abstractions can arise from domain-general cognitive mechanisms.

1. Introduction

Categories serve as a functional foundation for linguistic productivity. They allow words to be combined, to generalize across items, and to form abstractions that support flexible communication. Their importance for language learning and evolution has been widely recognized (Zhang, Shi, & Pleyer, 2025; Hurford, 2007; Bybee, 2010), and comparative research suggests that category learning may display evolutionary continuity across species. Studies show that many animals can group stimuli and respond to regularities (e.g. Wasserman & Zentall, 2012; Pepperberg, 1999; Smith, Zakrzewski, Johnson, Valleau, & Church, 2016). Yet it remains unclear whether animals form the kind of combinatorial categories that underpin human language. Human linguistic categories appear not only to support discrimination but to make an otherwise unlearnable system tractable, allowing generalization and flexible recombination within a limited working memory.

Generalization itself is a fundamental property of all learning organisms (Hull, 1943; Ghirlanda & Enquist, 2003). We suggest that the specific linguistic form of category learning emerges when generalization operates over sequences that are chunked. In other words, we add to the general capacity for generalization the uniquely human capacity for sequence representation.

We thus explore the possibility that linguistic categories may themselves emerge from more fundamental cognitive capacities – sequence representation and chunking – without being predefined. According to *the sequence hypothesis*, humans are uniquely sensitive to exact sequential order (Ghirlanda, Lind, & Enquist, 2017; Enquist, Ghirlanda, & Lind, 2023; Lind, Vinken, Jonsson, Ghirlanda, & Enquist, 2023; Lind & Jon-And, 2025b, 2025a; Jon-And, Jonsson, Lind, Ghirlanda, & Enquist, 2023), and the ability to faithfully represent and chunk ordered information is thought to underpin higher cognition, including language. If a system can generalize over these capacities across multiple levels, both hierarchical structure and flexible categories may emerge naturally.

It was shown in Jon-And & Michaud (2024b) that hierarchical structure can indeed emerge from such minimal cognitive mechanisms. Using a model based on sequence memory and chunking, and driven by an error-correction algorithm (Rescorla & Wagner, 1972; Sutton & Barto, 2018) that exploits the hierarchical structure of the states, the model was able to develop nested compositional patterns supporting a range of sentence types including mono- and ditransitive verbs, complex nouns phrases, and subordinate clauses. A first attempt to include a mechanism for generalization was developed in Jon-And & Michaud (2024a), where combinatorial categories were demonstrated to emerge for very simple two-word sentences. The generalization process was based on Lambek's syntactic type system (Lambek, 1958) with the central modification that the system was made *evolutionary* in the sense that the syntactic types were not predefined, but invented, replicated, and selected during processing.

Here, we extend this approach to show that emergent categories can arise flexibly across a broad range of constructions, including intransitive, monotransitive, and ditransitive verbs, and subordinate clauses. The emergent categories behave analogously to grammatical classes: they encode how elements can be combined and recombined in higher-order abstractions, supporting learning. The model does not explicitly represent semantic, pragmatic, or communicative information. What emerges are formal combinatorial representations that support sequence processing, although the task of identifying meaningful units implicitly reflects pressures relevant for communication. Crucially, the categories emerge not from language-specific biases but from domain-general cognitive properties of sequence representation and chunking, formalized through a self-organizing, evolving type framework.

2. The learning task

During language learning words are perceived in a stream, and the identification of meaningful units (in our case sentences) is a real task, which grounds our model in cognitively plausible assumptions. Following Sutton & Barto (2018), we define the task as a Markov Decision Process (MDP). MDPs provide a useful framework for studying learning from interactions with an environment to achieve a goal.

In our case, the goal of a learner is to identify sentences in a stream of words constituting the environment. Word frequency is inversely proportional to their rank, approximating distributions in natural languages (Zipf, 1932). Any hint as to the beginning of sentences, for example capital letters or punctuation are removed from the list of words. For example, a sequence of sentences

The cat chases the dog. The man loves his girlfriend. The sun shines.

will be transformed into

the cat chases the dog the man loves his girlfriend the sun shines

and the goal is to identify sentences boundaries. The information about sentence boundaries is thus masked and used to drive the agent's learning.

In a generic MDP, the agent starts in a state, takes actions to move to a new state, and gets rewards (positive or negative) to drive the learning process. In our model, the possible states are encoded as pairs of structured *chunks* of the word stream. A chunk refers to a sequence of words and their associated types, and an associated binary tree with the words-types pairs as leaves. By default, the second and most recently perceived chunk always consists of a single word-type pair. An example state is illustrated in the left part of Figure 1.

From a given state, and following an incremental processing of input akin to a chunk-and-pass mechanism (Christiansen & Chater, 2016), the learner needs to decide whether to insert the second element in the currently analyzed structure or to place a boundary between the two chunks. The possible resulting states corresponding to our example are displayed in Figure 1. Cases (1), (2), and (3) correspond to chunking actions inserting the second element at different levels in the tree and case (4) corresponds to boundary placement. Positive rewards are given whenever an identified sentence is correct and negative rewards are given otherwise. In this example, only case (1) is compatible with the assigned types since the list of types $[a, [a\backslash s/a, [a, a\backslash a/a]]]$ reduces to s/a. Placing a boundary is also incompatible with the assigned types because while the first element correctly reduces to a sentence, the second element expects to be preceded by a.

3. Inventing grammar

The learning model used in this paper builds on the models presented in (Jon-And & Michaud, 2024b, 2024a) by developing a more extensive generalization procedure using Lambek's syntactic types (Lambek, 1958). The main challenge of such a model is to invent an efficient type system starting with a single type s representing a well-formed sentence, functioning as a marker of successful identification of a meaningful unit. In this section, we explain our strategy to bootstrap such a type system using the sentence segmentation task describe above.

While the model developed by Jon-And & Michaud (2024b) is able to segment a word stream into sentences, every single sentence needs to be learned independently, which is not realistic in real language learning. However, the capacity to

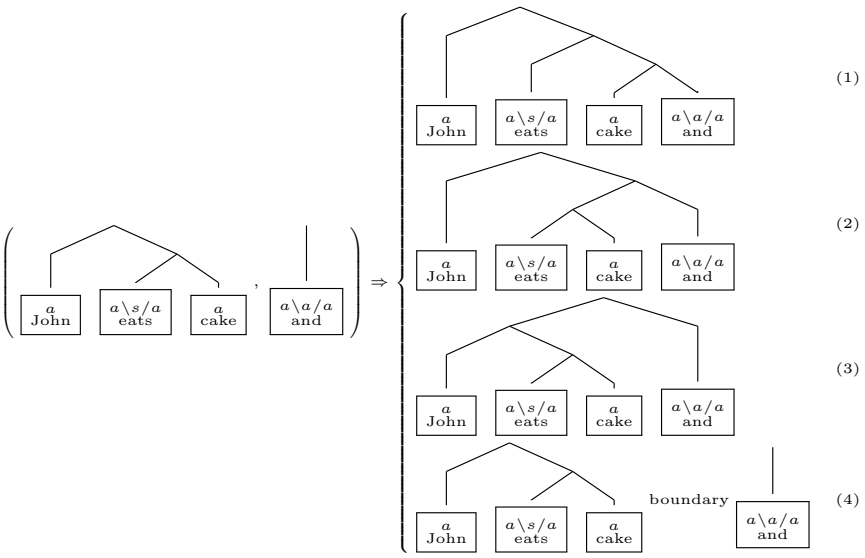

Figure 1. Possible decisions in a given state.

identify well-formed sentences can be used to bootstrap a type system. To do so, we associate the type s to well-formed sentences and associate compatible types to the leaves of the tree by introducing left and right expectations. For example, a two-word sentence *John eats* can be initially typed either $[s, s\backslash s]$ or $[s/s, s]$.

The emerging types then guide the MDP illustrated in Figure 1 by supporting decisions compatible with the assigned types. When the type assignment suggests boundary placement, word-type associations are reinforced. Types with negative associations with words block their assignment and trigger the emergence of new types, enabling the system to introduce new primitives when needed. The new primitives can be used to change the expectations of other types to enforce consistency. In addition, types expecting forward or backward can also assign types to yet untyped words. Through this process unknown words can be associated to known types, allowing the sentence processing to continue despite the unknown element, similarly to what is required in the so-called *wug* test (Berko, 1958), in which an invented animal called a *wug* is used to test if children can generalize and correctly predict the plural form of the word *wug* to be *wugs*.

4. Results and discussion

We use the learning model without a type system developed by Jon-And and Michaud (2024b) as a baseline for evaluating our model with emergent types. Here, we tested the model on two artificial languages, the first one includ-

ing monotransitive and ditransitive constructions, the second one adding relative clauses. The artificial languages encode hierarchical and argument-structure regularities, reflecting the fact that natural language is culturally structured. The emergent types should therefore be understood not as creating categories from scratch, but as rediscovering regularities in the input. In both languages, learners with emergent types outperformed non-type learners, with faster convergence and higher accuracy in identifying sentences (Fig. 2). The same functional categories consistently emerged across languages (Table 1). Nouns were assigned a primitive type, monotransitive and ditransitive verbs types encoding the number of expected noun arguments, and relative pronouns a type expecting a sentence to its left, yielding a noun-like category. Relative pronouns expect a sentence to its left, reflecting the structure of the simplified language used here, where relative clauses only occur at the end of sentences. If relative clauses are added after nouns in any position, a more complex and less uniform grammar emerges. The typing of relative pronouns in this simplified language shows that the emergent grammatical categories are adaptive and assumes the simplest possible form. Overall, the emergent types efficiently capture the combinatorial structure of each construction and mirror properties of grammatical classes in the information they encode.

The type learner performs well when both vocabulary size and structural complexity increase, the system scaling efficiently when combinatorial demands rise. An interesting pattern is the initially similar learning curves of type and non-type learners, reflecting that both initially rely on frequent substantial chunks. The type learner is only able to induce a functional grammatical system after exposure to sufficient data. This early chunk-based stage parallels findings in natural language acquisition, where frequently occurring multi-word sequences provide the foundation for later abstraction and productivity (Tomasello, 2003; Bybee, 2010). Recurrent chunks act as scaffolds: they enable reliable early performance and later serve as the basis for category-based generalization. Once the type learner starts to induce functional categories, they reproduce efficiently by assigning new types to surrounding elements, or adjusting type assignments to be compatible with each other. This speeds up learning drastically, as can be seen in Fig. 2.

The emergent types bear resemblance to constructions in the Construction Grammar framework. Constructions are learned form–meaning pairings that range from concrete expressions to abstract patterns with open slots (Goldberg, 2007). Our types are abstractions over recurrent combinatorial patterns, encoding how elements can be combined while leaving open positions for variable fillers. Primitives are generalized based on distributional substitutability – the words that can fill a given slot – mirroring how lexical classes arise in usage-based accounts. In this sense, while lacking the meaning side, our evolving types provide a formal instantiation of usage-based emergence of constructions (Croft, 2001).

Our findings also connect to work on the transition from continuous to symbolic representation of information in cognitive science (Smolensky, 1988; Mar-

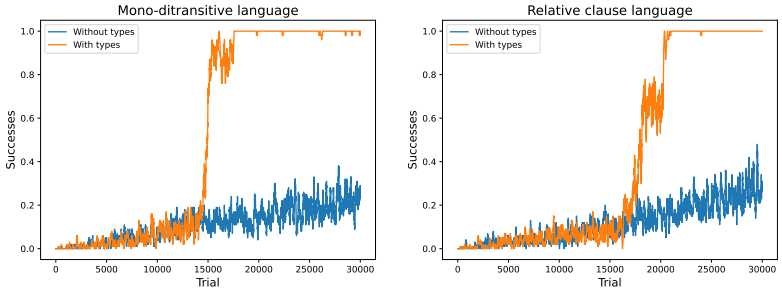

Figure 2. Left: Learning curves for a language with mono- and ditransitive verbs (20 nouns, 10 mono, 10 ditransitive verbs) with and without types. Right: Learning curves for a language also including relative clauses (40 nouns, 10 mono, 2 ditransitive verbs, and 2 relative pronouns).

cus, 2001) or neuro-symbolic AI (Smolensky, McCoy, Fernandez, Goldrick, & Gao, 2022; Zhang & Sheng, 2024; Liang, Wang, & Tong, 2025; Colelough & Regli, 2025), which, typically relies on explicit symbolic frames or hybrid architectures. Even when symbolic behavior is learned, as in grounded concept models (Nevens, van Eecke, & Beuls, 2020), the symbols themselves are predefined representational targets rather than emergent solutions to learning constraints. In contrast, our model shows how symbolic-like categories can arise *spontaneously* as an adaptive response to the combinatorial challenges of language learning.

Table 1. Types most strongly associated to each respective word class, consistent over both languages.

Noun	Monotransitive Verb	Ditransitive Verb	Relative Pronoun
a	$a\backslash s/a$	$a\backslash s/a/a$	$s\backslash a$

Although grammatical categories may not constitute symbols in the traditional sense, since they are learned implicitly and not represented as discrete tokens by language users, they nevertheless perform a symbolic-like function. As the number of possible sequences grows, learning without categories quickly becomes intractable. Categories arise in our system precisely when they are needed to make learning feasible, compressing recurrent combinatorial patterns into reusable abstractions. Our model illustrates how structured, symbol-like representations can evolve from usage-based learning, with minimal and cognitively plausible assumptions about sequence representation and composition.

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